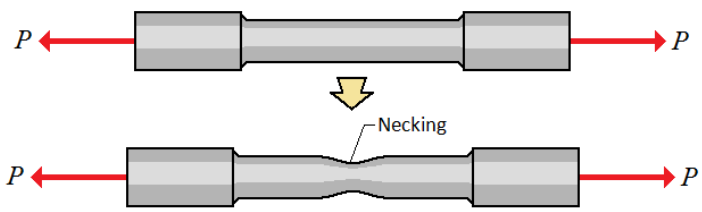
October 7, 2022

APS110 – Lecture 13

See hand written notes for drawings.

Strength → a specific value of stress

- Stress-strain curve
  - Peak of curve → UTS (highest stress recorded)
    - At this point the sample develops **localized plastic deformation** (necking)
    - Localized reduction in dimensions of the sample
    - When a metal fails, it will occur at this necked region



  - Where is the yield strength?
    - Usually close to the proportional limit (near the end of the linear region)
    - But, it's usually a little after that → there is a non-linear elastic region of the curve
    - It's difficult to determine exactly where this is, therefore we need a convention
      - **0.2% limit – strain = 0.002**
      - We take strain at this value and we draw a straight line with the same slope as the curve (E)
      - At some point this line will intersect the curve, and this point is where we say our **yield strength** is

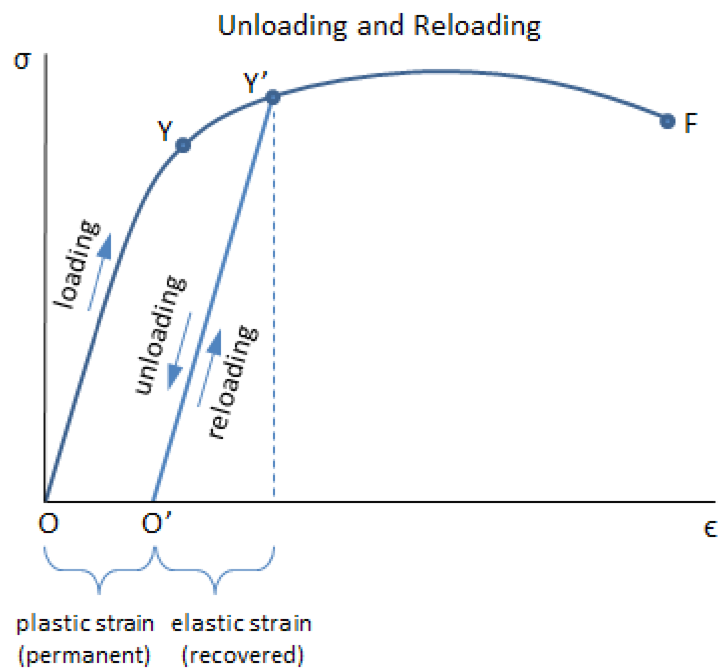
Typical Metal

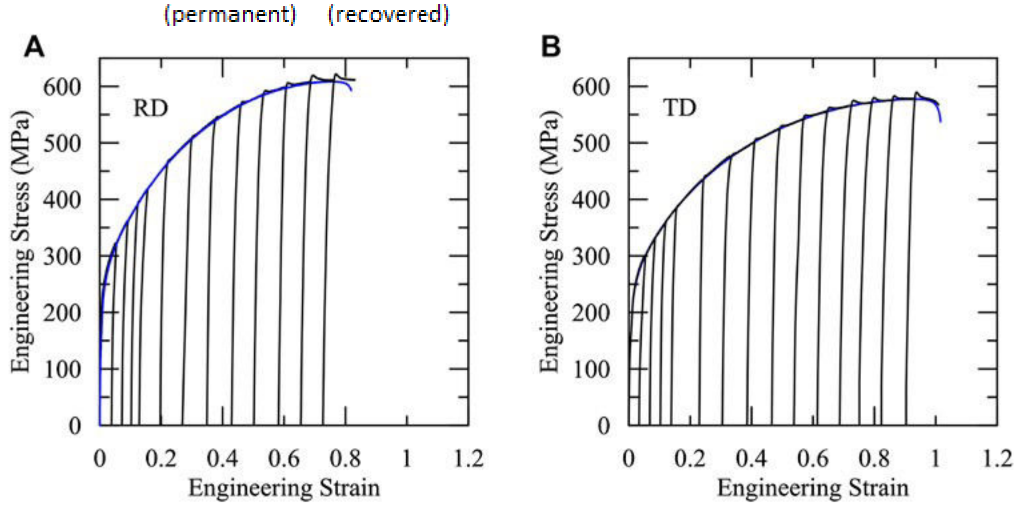
- Stress-strain curve
  - Uniform plastic → uniformly distributed along reduced section
    - From end of elastic to peak of curve (UTS point)
  - Non-uniform plastic, or localized → deformation that is localized to a small region of the reduced section
  - We typically don't consider the non-linear elastic region in practice, we just assume the curve is linear until the yield strength

Unloading, then reloading

- We can strengthen some metals by deforming them elastically, then unloading, then reloading, then unloading, etc… then each time we unload then reload the material, the yield strength increases → the material will get stronger and stronger
- The consequences of this is the material will lose ductility → we are reducing the capacity for plastic deformation each time we do this and so the material will get stronger but it will also get more brittle

October 7, 2022





- Notice the curve keeps increasing until fracture (it might plateau at some point)
- **Strain hardening →** strengthening of a material after plastic deformation

Plastic deformation at the atomic level

- The diagram we see here is accurate if we consider that the model on the left is the 'start' and the one on the right is the 'end' → we are missing the intermediate here
  - What is the mechanism?
- The mechanism for plastic deformation involves a step-by-step breaking and reforming of bonds
  - The whole plane of atoms doesn't not move all at once, there is an incremental breaking and moving of bonds and atoms